

Adaptive Process Noise Estimation in Sim2Real Frameworks: A Comparative Analysis Across Nonlinear Kalman Filters

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Abstract—Accurate underwater navigation is challenging due to the unavailability of GNSS signals and time-varying uncertainties in inertial and Doppler velocity log measurements. Several works have investigated the contribution of neural-aided adaptive process noise covariance to positioning accuracy, yet without a common ground for comparison and analysis. To bridge this gap, we benchmark three nonlinear Kalman filter variants under an adaptive process noise estimation mechanism designed using a Sim2Real training approach. Through this comparative benchmarking framework and using 80.8 minutes of real-world autonomous underwater vehicles trajectories, we demonstrate that the Sim2Real adaptive invariant Kalman filter outperforms the baseline Sim2Real adaptive extended Kalman filter by 57.2%. These results provide practical guidance for selecting a nonlinear filtering approach in Sim2Real strategies, and highlight the benefit of combining adaptive process noise estimation with invariant filtering for underwater navigation.

Index Terms—Kalman filter, Sensor fusion, Adaptive process noise, State estimation, Lie group.

I. INTRODUCTION

Accurate underwater navigation is challenging because global navigation satellite system (GNSS) signals are unavailable below the water surface. Autonomous underwater vehicles (AUVs) must therefore rely on onboard sensor fusion, commonly combining inertial measurements with Doppler velocity log (DVL) observations, to estimate their position, velocity, and orientation.

Nonlinear Kalman filters, including the extended Kalman filter (EKF) [1], [2], unscented Kalman filter (UKF) [3], [4], and right-invariant Kalman filter (RI-UKF) [5]–[7], provide established frameworks for inertial/DVL sensor fusion [8]–[12]. Their performance, however, depends strongly on

the process noise covariance matrix, which is commonly fixed and manually tuned. Because AUV dynamics, sensor characteristics, and environmental disturbances vary over time, a fixed process noise covariance may not adequately represent the uncertainty of the navigation model.

Adaptive Kalman filtering has been investigated extensively for INS/DVL navigation under uncertain and time-varying noise conditions. Existing methods include innovation-based covariance adaptation [13]–[16], recursive noise estimation [17], online system noise covariance estimation [18], [19], adaptive nonlinear filtering [20], and variational Bayesian covariance estimation [21]. Robust adaptive filters have also been developed for complex underwater environments [22]. In parallel, invariant error formulations have been applied to SINS/DVL integration, including adaptive and robust variants designed to address inaccurate process and measurement noise covariances [23], [24].

More recently, neural-aided approaches have been used to estimate time-varying inertial-noise parameters and adapt the process noise covariance online [25]–[28]. However, existing studies generally employ different filtering formulations, datasets, sensor configurations, and adaptation mechanisms. Consequently, their reported improvements do not provide a common basis for determining how the underlying nonlinear filtering formulation influences the benefit obtained from adaptive process noise covariance estimation.

This paper provides a common-ground, real-world benchmark of the EKF, UKF, and RI-UKF and their neural-aided variants under identical conditions. All three variants employ the Sim2Real adaptive mechanism

introduced in [27], which combines neural inertial-noise estimation with innovation-based covariance adaptation.

The Sim2Real approach utilizes domain randomization on simulated trajectories to train a noise-estimating neural network, enabling direct online deployment on unlabelled real-world data. Our evaluation reveals that the Neural-Aided RI-IKF leverages this adaptation most effectively, reducing mean position error by 42.4% and 57.2% compared to the neural-aided UKF and EKF, respectively, thereby demonstrating a strong synergy between data-driven covariance estimation and Lie-group formulations.

The remainder of this paper is organized as follows. Section II introduces the EKF, UKF, and RI-IKF formulations. Section III describes the common adaptive covariance-estimation mechanism. Section IV presents the benchmark dataset, evaluated filtering approaches, and experimental results. Section V concludes the paper.

II. PROBLEM FORMULATION

The inertial navigation equations expressed in the earth centered earth frame (ECEF) are [29].

$$\begin{aligned}\dot{\mathbf{p}}_{eb}^e &= \mathbf{v}_{eb}^e \\ \dot{\mathbf{v}}_{eb}^e &= \mathbf{C}_b^e \mathbf{f}_{ib}^b - 2(\boldsymbol{\omega}_{ie}^e)^\wedge \mathbf{v}_{eb}^e + \mathbf{g}_{ib}^e \\ \dot{\mathbf{C}}_b^e &= \mathbf{C}_b^e (\boldsymbol{\omega}_{ib}^b)^\wedge - (\boldsymbol{\omega}_{ie}^e)^\wedge \mathbf{C}_b^e\end{aligned}\quad (1)$$

where the $\mathbf{p}_{eb}^e$ and $\mathbf{v}_{eb}^e$ are the position and velocity, respectively, of the body frame relative to ECEF frame resolved in ECEF frame. $\boldsymbol{\omega}_{ib}^b$ and $\mathbf{f}_{ib}^b$ are the angular velocity and the specific force vectors of body with respect to the inertial frame resolved in body frame, respectively, and $\mathbf{C}_b^e$ is the rotation matrix from body frame to ECEF frame.

A. Extend Kalman filter

The error-state vector and the corresponding process-noise vector are defined as [1], [8]

$$\delta \mathbf{x}_{\text{EKF}} = \begin{bmatrix} \delta \phi^b \\ \delta \mathbf{v}_{eb}^e \\ \delta \mathbf{p}_{eb}^e \\ \delta \mathbf{b}_g \\ \delta \mathbf{b}_a \end{bmatrix} \in \mathbb{R}^{15} \quad \mathbf{w} = \begin{bmatrix} \mathbf{n}_g \\ \mathbf{n}_a \\ \mathbf{n}_{b_g} \\ \mathbf{n}_{b_a} \end{bmatrix} \in \mathbb{R}^{12} \quad (2)$$

where $\mathbf{w} \in \mathbb{R}^{12}$ consists of the gyroscope noise, accelerometer noise, bias gyroscope noise and bias accelerometer noise, all modeled as zero mean white Gaussian noise.

The error state dynamics comply with the following continuous differential equation

$$\delta \dot{\mathbf{x}}_{\text{EKF}} = \mathbf{F}_{\text{EKF}} \delta \mathbf{x}_{\text{EKF}} + \mathbf{G}_{\text{EKF}} \mathbf{w} \quad (3)$$

where the system matrix

$$\mathbf{F}_{\text{EKF}} = \begin{bmatrix} -\boldsymbol{\Omega}_{ie}^e & \mathbf{0} & \mathbf{0} & -\mathbf{C}_b^e & \mathbf{0} \\ -(\mathbf{C}_b^e \mathbf{f}_{ib}^b)^\wedge & -2\boldsymbol{\Omega}_{ie}^{e,\wedge} & \mathbf{g} - (\boldsymbol{\Omega}_{ie}^e)^2 & \mathbf{0} & -\mathbf{C}_b^e \\ \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad (4)$$

where $\mathbf{g}$ is the local gravity vector and $\boldsymbol{\Omega}_{ie}^e$ is the skew-symmetric matrix of the Earth's rotation rate. The noise shaping matrix is

$$\mathbf{G}_{\text{EKF}} = \begin{bmatrix} -\mathbf{C}_b^e & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -\mathbf{C}_b^e & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & -\mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & -\mathbf{I}_3 \end{bmatrix}, \quad (5)$$

The measurement residual of the DVL update is:

$$\delta \mathbf{y} = \mathbf{H}_{\text{EKF}} \delta \mathbf{x}_{\text{EKF}} \quad (6)$$

where the measurement model matrix is

$$\mathbf{H}_{\text{DVL}} = [\mathbf{C}_e^b (\mathbf{v}_{eb}^e)^\wedge \quad \mathbf{C}_e^b \quad \mathbf{0}_{3 \times 9}] \quad (7)$$

B. Invariant Kalman filter

The right-invariant error-state model is formulated in ECEF frame following [5] [30] by convention

$$\eta^R = \hat{\mathbf{X}} \mathbf{X}^{-1} = \exp((\xi^R)^\wedge) \quad (8)$$

where

$$\exp(A) = \sum_{k=0}^{\infty} \frac{A^k}{k!} \quad (9)$$

is the mapping from the Lie algebra $\mathfrak{se}_2(3)$ to the Lie group $SE_2(3)$, the Lie discrepancy is

$$(\xi^R)^\wedge = \begin{bmatrix} 0 & -\xi_3^R & \xi_2^R & \xi_4^R & \xi_7^R \\ \xi_3^R & 0 & -\xi_1^R & \xi_5^R & \xi_8^R \\ -\xi_2^R & \xi_1^R & 0 & \xi_6^R & \xi_9^R \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \in \mathfrak{se}_2(3), \quad (10)$$

$\mathbf{X}$ is the ground truth (GT) state encoded in Lie group defined by

$$\mathbf{X} = \begin{bmatrix} \mathbf{C}_b^e & \mathbf{v}_{eb}^e & \mathbf{p}_{eb}^e \\ 0_{1 \times 3} & 1 & 0 \\ 0_{1 \times 3} & 0 & 1 \end{bmatrix} \in SE_2(3), \quad (11)$$

$\hat{\mathbf{X}}$ is the estimated state.

Given that, the right invariant error is

$$\eta^R = \begin{bmatrix} \hat{\mathbf{C}}_b^e \mathbf{C}_e^b & \hat{\mathbf{v}}_{eb}^e - \hat{\mathbf{C}}_b^e \mathbf{v}_{eb}^e & \hat{\mathbf{p}}_{eb}^e - \hat{\mathbf{C}}_b^e \mathbf{p}_{eb}^e \\ 0_{1 \times 3} & 1 & 0 \\ 0_{1 \times 3} & 0 & 1 \end{bmatrix} \quad (12)$$

with

$$\hat{\mathbf{C}}_b^e \mathbf{C}_e^b = \exp(\phi^{e,\wedge}) \approx \mathbf{I} + \phi^{e,\wedge} \quad (13)$$

the vectorized right invariant error is:

$$\delta \mathbf{x}_{\text{IEKF}} = \log(\eta^R)^\vee \quad (14)$$

where $\vee$ is the inverse of (10) and $\log$ is the inverse of (9)

The right invariant error state dynamics are

$$\delta \dot{\mathbf{x}}_{\text{IEKF}} = \mathbf{F}_{\text{IEKF}} \delta \mathbf{x}_{\text{IEKF}} + \mathbf{G}_{\text{IEKF}} \mathbf{w} \quad (15)$$

where $\mathbf{w} \in \mathbb{R}^{12}$ is define in (2) and the system matrix is given by:

$$\mathbf{F}_{\text{IEKF}} = \begin{bmatrix} -\Omega_{ie}^e & \mathbf{0} & \mathbf{0} & \mathbf{C}_b^e & \mathbf{0} \\ -\mathbf{v}^\wedge \Omega_{ie}^e + \mathbf{g}^\wedge & -2\Omega_{ie}^e & \mathbf{0} & \mathbf{v}^\wedge \mathbf{C}_b^e & \mathbf{C}_b^e \\ \mathbf{0} & \mathbf{I}_3 & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (16)$$

The noise shaping matrix is

$$\mathbf{G}_{\text{IEKF}} = \begin{bmatrix} \mathbf{C}_b^e & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{v}^\wedge \mathbf{C}_b^e & \mathbf{C}_b^e & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_3 \end{bmatrix} \quad (17)$$

where the measurement model of the RI-IKF is

$$\mathbf{H}_{\text{IEKF}} = [\mathbf{0}_{3 \times 3} \quad -\mathbf{I}_3 \quad \mathbf{0}_{3 \times 9}] \quad (18)$$

For calculating the Kalman gain, posterior state covariance, posterior state read [30] or [27].

C. Unscented Kalman filter

Consider the initialized state vector $\mathbf{x}_0$ with a known mean and covariance:

$$\hat{\mathbf{x}}_0 = \mathbb{E}[\mathbf{x}_0], \quad (19)$$

$$\mathbf{P}_0 = \mathbb{E}[(\mathbf{x}_0 - \hat{\mathbf{x}}_0)(\mathbf{x}_0 - \hat{\mathbf{x}}_0)^\top]. \quad (20)$$

The weights for computing the mean (m upper case) and covariance (c upper case) are defined as:

$$w_0^{(m)} = \frac{\lambda}{n + \lambda}, \quad (21)$$

$$w_0^{(c)} = \frac{\lambda}{n + \lambda} + (1 + \alpha^2 + \beta), \quad (22)$$

$$w_i^{(m)} = w_i^{(c)} = \frac{1}{2(n + \lambda)}, \quad i = 1, \dots, 2n, \quad (23)$$

where n is the dimension of the state vector and

$$\lambda = \alpha^2(n + \kappa) - n \quad (24)$$

The parameters α , β , and κ control the spread and weighting of the sigma points and are addressed as hyperparameters. Typically, $\alpha = 10^{-3}$, $\kappa = 0$, and $\beta = 2$ for Gaussian distributions.

1) *Sigma Points*: The sigma points are constructed as

$$\chi_{0,k} = \hat{\mathbf{x}}_k, \quad i = 0 \quad (25)$$

$$\chi_{i,k} = \hat{\mathbf{x}}_k + \left[\sqrt{(n + \lambda)\mathbf{P}_k} \right]_i, \quad i = 1, \dots, n, \quad (26)$$

$$\chi_{i,k} = \hat{\mathbf{x}}_k - \left[\sqrt{(n + \lambda)\mathbf{P}_k} \right]_{i-n}, \quad i = n + 1, \dots, 2n, \quad (27)$$

where $\sqrt{\mathbf{P}_k}$ denotes the Cholesky decomposition of the covariance matrix. Each sigma point is propagated through the nonlinear model:

$$\chi_{i,k+1|k} = f(\chi_{i,k}), \quad i = 0, \dots, 2n. \quad (28)$$

The predicted state mean and covariance are then computed as

$$\hat{\mathbf{x}}_{k+1|k}^- = \sum_{i=0}^{2n} w_i^{(m)} \chi_{i,k+1|k} \quad (29)$$

$$\mathbf{P}_{k+1|k}^- = \sum_{i=0}^{2n} w_i^{(c)} (\chi_{i,k+1|k} - \hat{\mathbf{x}}_{k+1|k}^-)(\chi_{i,k+1|k} - \hat{\mathbf{x}}_{k+1|k}^-)^\top + \mathbf{Q}_{k+1}. \quad (30)$$

where $\mathbf{Q}_{k+1}$ is the covariance process noise matrix in time stamp $k+1$.

2) *Update step*: A new set of sigma points is generated from the predicted mean and covariance and propagated through the measurement function:

$$\mathcal{Z}_{i,k+1|k} = h(\chi_{i,k+1|k}), \quad i = 0, \dots, 2n. \quad (31)$$

And the measurement model of the DVL is:

$$\mathcal{Z}_{i,k+1|k} = h(\chi_{i,k+1|k}) \quad (32)$$

$$= \mathbf{C}_b^n (\mathbf{I} - \phi^{n\wedge}) [\mathbf{v}^n + \delta \hat{\mathbf{v}}_n^i] - \mathbf{C}_b^n \mathbf{v}^n \quad (33)$$

$$i = 0, \dots, 2n \quad (34)$$

The predicted measurement mean and covariance are computed as:

$$\hat{\mathbf{z}}_{k+1|k} = \sum_{i=0}^{2n} w_i^{(m)} \mathcal{Z}_{i,k+1|k} \quad (35)$$

$$\mathbf{S}_{k+1} = \sum_{i=0}^{2n} w_i^{(c)} (\mathcal{Z}_{i,k+1|k} - \hat{\mathbf{z}}_{k+1|k})(\mathcal{Z}_{i,k+1|k} - \hat{\mathbf{z}}_{k+1|k})^\top + \mathbf{R}_{k+1}. \quad (36)$$

The cross-covariance matrix is define by:

$$\mathbf{P}_{xz,k+1} = \sum_{i=0}^{2n} w_i^{(c)} (\chi_{i,k+1|k} - \hat{\mathbf{x}}_{k+1|k})(\mathcal{Z}_{i,k+1|k} - \hat{\mathbf{z}}_{k+1|k})^\top \quad (37)$$

The Kalman gain is computed by using (36) and (37):

$$\mathbf{K}_{k+1} = \mathbf{P}_{xz,k+1} \mathbf{S}_{k+1}^{-1} \quad (38)$$

Finally, the state mean and covariance are updated using the measurement $\mathbf{z}_{k+1}$ and the Kalman gain (38):

$$\hat{\mathbf{x}}_{k+1}^+ = \hat{\mathbf{x}}_{k+1|k}^- + \mathbf{K}_{k+1}(\mathbf{z}_{k+1} - \hat{\mathbf{z}}_{k+1|k}) \quad (39)$$

$$\mathbf{P}_{k+1}^+ = \mathbf{P}_{k+1|k}^- - \mathbf{K}_{k+1} \mathbf{S}_{k+1} \mathbf{K}_{k+1}^\top \quad (40)$$

III. BENCHMARKING METHODOLOGY

To provide a common basis for comparison, the same adaptive process noise covariance-estimation mechanism is applied to the EKF, UKF, and RI-IKF. The mechanism, based on the approach introduced in [27], combines two sources of information: neural estimates of time-varying inertial sensor noise and covariance information derived from the filter innovations. The neural component is shared by all three filtering formulations, including its architecture,

trained weights, input window, and output parameters. The innovation-based component is implemented according to the error representation of each filter. This common configuration enables the effect of the underlying filtering formulation to be evaluated independently of the neural model and adaptation settings.

The neural network is trained using simulated inertial datasets generated from trajectories with varying motion profiles and sensor-noise levels, following the Sim2Real methodology introduced in [27]. In this approach, domain randomization is used to generate diverse simulated trajectories with varying motion profiles, initial conditions, and accelerometer and gyroscope noise levels. This enables the network to learn the relationship between short windows of raw inertial measurements and the corresponding process-noise parameters, without requiring labeled real-world data. Following the Sim2Real method, after training on the simulative data, the network is evaluated on real recorded dataset. The resulting inertial-noise estimates are used to construct a time-varying process noise covariance matrix for each filtering formulation. The complete covariance-estimation mechanism is illustrated in Fig. 1.

The adaptive process noise covariance consists of two components. The first is a data-driven component obtained from a window of recent inertial measurements. The second is a model-based component obtained from a window of filter innovations. The EKF and UKF use the Euclidean innovation-based formulation in [13], whereas the RI-IKF uses the Lie-group-based formulation described in [27].

A. Neural-aided inertial learning approach

The neural-learning-based approach in [27] maps a window of recent inertial measurements to the gyroscope and accelerometer process-noise parameters:

$$\mathbf{q} = f_{\theta}(\mathcal{W}) = \begin{bmatrix} \mathbf{q}_g \\ \mathbf{q}_a \end{bmatrix} \in \mathbb{R}^6, \quad (41)$$

where f_{θ} denotes the neural network, $\mathcal{W}$ is a window of recent inertial measurements, and $\mathbf{q}_g, \mathbf{q}_a \in \mathbb{R}^3$ contain the predicted process-noise variances of the gyroscope and accelerometer, respectively. The continuous-time process-noise covariance matrix is then defined as

$$\mathbf{Q}_{c,\text{neural}} = \text{diag}(\mathbf{q}_g, \mathbf{q}_a, \mathbf{q}_{bg}, \mathbf{q}_{ba}) \in \mathbb{R}^{12 \times 12}, \quad (42)$$

where $\mathbf{q}_{bg}, \mathbf{q}_{ba} \in \mathbb{R}^3$ represent the fixed gyroscope-bias and accelerometer-bias process-noise variances, respectively.

B. Adaptive innovation model-based approach EKF/UKF

The EKF and UKF require Euclidean innovation which is the difference between the DVL measurement and the estimated velocity in body frame which is

$$\nu_{\text{innov}} = v_{eb}^b - \hat{v}_{eb}^b \quad (43)$$

where v_{eb}^b is the velocity expressed in the body frame as measured by the DVL and $\hat{v}_{eb}^b$ is the estimated velocity expressed in the body.

The adaptive innovation-based process noise in steady state is given by [13]

$$\mathbf{Q}_{\text{innov}} = \frac{1}{N} \sum_{n=0}^N K_n \nu_n \nu_n^T K_n^T \quad (44)$$

where $\mathbf{K}_n$ is the Kalman gain at step n given by the EKF or the UKF, and ν_n is the innovation.

C. Adaptive innovation model-based approach RI-IKF

As shown in [27] the adaptive innovation-based process noise covariance that is derived from Lie innovation is

$$\mathbf{Q}_{\text{innov}} = \frac{1}{N} \sum_{n=1}^N \log(\hat{\mathbf{X}} \mathbf{X}^{-1})^{\vee} \log(\hat{\mathbf{X}} \mathbf{X}^{-1})^{\vee, T} \quad (45)$$

where $\mathbf{X}$ is the posterior state and $\hat{\mathbf{X}}$ is the prior state and the mapping $(\cdot)^{\vee}$ is given by the inverse of (10)

D. Combined Process Noise Covariance

Ultimately, The discrete-time process-noise covariance used by all Kalman filter variants is obtained by averaging the neural-network-based covariance and the adaptive innovation-based covariance:

$$\mathbf{Q}_d = \frac{1}{2} (\mathbf{G} \mathbf{Q}_{c,\text{neural}} \mathbf{G}^T + \mathbf{Q}_{d,\text{innovation}}) \quad (46)$$

where the neural process noise covariance $\mathbf{Q}_{c,\text{neural}}$ is obtained by the same neural network with the same weights in all neural-aided Kalman filters. The innovation based covariance in neural-aided EKF and neural-aided UKF is given by (44) and the innovation based covariance $\mathbf{Q}_{d,\text{innovation}}$ in neural-aided RI-IKF is given by (45).

IV. ANALYSIS AND RESULTS

A. Test Dataset

The dataset used for the evaluation of our results (test dataset) was acquired by the Snapir AUV during maneuvers in the Mediterranean Sea [31]. with a total mission duration of 4,800 seconds. To ensure a consistent and controlled evaluation across all filtering frameworks, additional zero mean white Gaussian noise was injected into the inertial and DVL measurements. The accelerometer measurements were corrupted with zero mean white Gaussian noise with a standard deviation of $4 \times 10^{-3} \text{m/s}^2$, the gyroscope measurements with a standard deviation of $1 \times 10^{-4} \text{rad/s}$, and the DVL measurements with a standard deviation of $5 \times 10^{-3} \text{m/s}$. The same noise realization and noise levels were applied consistently to all evaluated filtering methods to ensure a fair comparison.

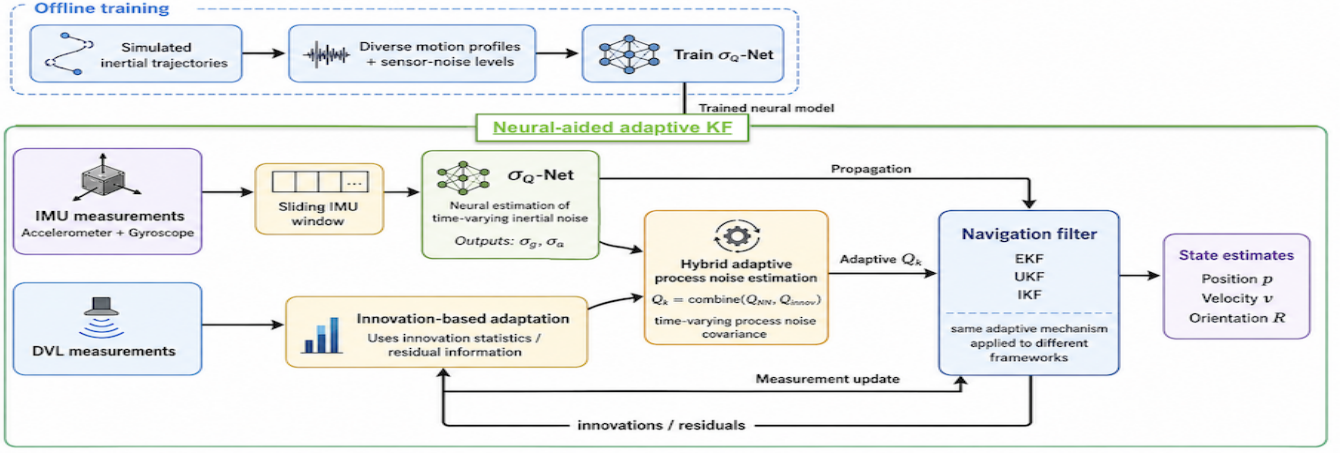


Fig. 1. Common adaptive covariance-estimation mechanism used in the benchmark. A neural network estimates time-varying inertial sensor noise from a sliding IMU window, while an innovation-based adaptive mechanism extracts covariance information from the filter residuals. The two estimates are combined to construct the adaptive process noise covariance matrix Q_k used by the Neural-Aided EKF, Neural-Aided UKF, and Neural-Aided RI-IKF.

B. Benchmarked Filtering Approaches

Six filtering approaches are included in the benchmark: three model-based nonlinear Kalman filters and their corresponding neural-aided adaptive variants. All approaches use the same inertial navigation dynamics in (1), IMU and DVL measurements, initial conditions, measurement noise covariance, and evaluation trajectories. The model-based approaches use a fixed process noise covariance matrix, whereas the neural-aided approaches use the adaptive process noise covariance defined in (46). The same neural network, trained weights, input window, and adaptation parameters are used for all three neural-aided approaches.

The six benchmarked approaches are:

- 1) **EKF**: The model-based extended Kalman filter based on the Euclidean error-state formulation defined in (2), using a fixed process noise covariance matrix.
- 2) **Neural-Aided EKF**: The same EKF formulation with the fixed process noise covariance replaced by the adaptive covariance in (46). Its innovation-based component is calculated using (44).
- 3) **UKF**: The model-based unscented Kalman filter II-C, in which the nonlinear state and measurement models are propagated using sigma points, with a fixed process noise covariance matrix.
- 4) **Neural-Aided UKF**: The same UKF formulation with the adaptive process noise covariance in (46). As in the Neural-Aided EKF, its innovation-based component is calculated using (44).
- 5) **RI-IKF**: The model-based right-invariant Kalman filter based on the Lie-group error representation in (8), using a fixed process noise covariance matrix.
- 6) **Neural-Aided RI-IKF**: The same RI-IKF formulation with the adaptive process noise covariance in (46). Its innovation-based component is calculated in the Lie algebra using (45).

This configuration provides a common ground for assessing

two effects: the performance differences among the EKF, UKF, and RI-IKF formulations and the improvement obtained by applying the same neural-aided adaptive covariance-estimation mechanism to each formulation.

C. Results

The EKF, UKF, and RI-IKF, together with their neural-aided variants, are evaluated against the post-processed ground-truth (GT) solution. Table I compares all six filtering approaches. In this benchmark, the position, velocity, and yaw root mean square errors are calculated relative to the post-processed GT solution. As shown in the table, the neural-aided variants consistently outperform their conventional model-based counterparts across all metrics. Notably, the Neural-Aided RI-IKF achieves the highest overall accuracy, yielding the lowest position error (7.61 m), velocity error (0.17 m/s), and yaw error (3.32°). Furthermore, while all neural frameworks demonstrate a significant reduction in position drift, the invariant formulation leverages the adaptive process noise most effectively. Figure 2 presents the PRMSE

TABLE I
AVERAGE NAVIGATION ERRORS OVER 12 AUV TRAJECTORIES.

Method	PRMSE [m]	VRMSE [m/s]	YRMSE [deg]
EKF	25.71	0.22	3.85
Neural EKF	17.80	0.18	3.44
UKF	27.51	0.25	4.01
Neural UKF	13.21	0.18	3.52
RI-IKF	21.64	0.24	3.42
Neural RI-IKF	7.61	0.17	3.32

comparison for all six methods. These results indicate that the invariant formulation obtains the largest positioning benefit from the common adaptive covariance-estimation mechanism. As visually demonstrated in the chart, the integration of the neural-aided mechanism yields a distinct, stepped reduction in positioning drift for each filter. While the Neural-Aided EKF and Neural-Aided UKF reduce position errors

by 30.8% and 52.0% respectively compared to their model-based baselines, the visualization highlights the superior adaptation of the Neural-Aided RI-IKF. It achieves a 64.8% error reduction compared to its conventional counterpart, vastly outperforming the relative gains observed in the other frameworks.

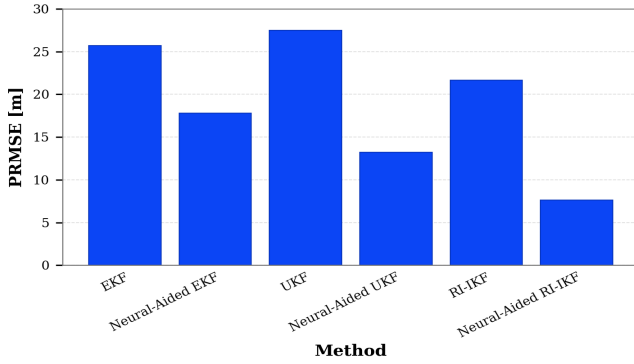


Fig. 2. PRMSE comparison of the EKF, UKF, and RI-IKF and their neural-aided variants.

V. CONCLUSION

This paper presented a common-ground benchmark of the EKF, UKF, RI-IKF, and their neural-aided adaptive variants using 12 real-world AUV trajectories totaling 4,800 seconds. All approaches were evaluated under identical sensor, noise, initialization, and adaptation conditions. Relative to their fixed-covariance counterparts, the Neural-Aided EKF, Neural-Aided UKF, and Neural-Aided RI-IKF reduced the position RMSE by 30.8%, 52.0%, and 64.8%, respectively. The Neural-Aided RI-IKF achieved the best performance, reducing the position RMSE by 42.4% relative to the Neural-Aided UKF and by 57.2% relative to the Neural-Aided EKF. These results demonstrate that the benefit of adaptive process noise estimation depends on the underlying filtering formulation and that, under the evaluated of AUV navigation, the invariant formulation benefits most from the common adaptation mechanism.

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